

Assignment 1: Imitation Learning

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Collaborators: Gemini (Guided Learning)

1 Behavioral Cloning (9.75 pt)

1.1 Part 2 (1.5 pt)

Table 1: Performance of Mean and Std. return over 2 trajectories on each environment.

Metric/Env	Ant-v2	Humanoid-v2	Walker2d-v2	Hopper-v2	HalfCheetah-v2
Mean	4713.65	10344.52	5566.85	3772.67	4205.78
Std.	12.20	20.98	9.24	1.95	83.04

1.2 Part 3 (5.25 pt)

Table 2: Evaluated on 5000 steps, **5 layers**, with **learning rate 7.6e-3**, others unchanged (detail could be view in Table ??) The discrepancy mainly lies in the difficulty of ‘balancing’. The Ant agent finds it easy to stay standing, and despite being random, most of the possible angle movements still allow it to move forward. While Hopper has a hard time dealing with physics, making it extremely difficult to stand, let alone move.

Env	Ant-v2		Hopper-v2	
Metric	Mean	Std.	Mean	Std.
Expert	4713.65	12.20	3772.67	1.95
BC	1462.01	458.68	215.10	83.54
Pct.(%)	31.02		5.70	

1.3 Part 4 (3 pt)

Originally, I noticed that there are no batch normalization layers in the network. So my initial thought was that in this case larger batches may not always lead to better results, as they tend to converge to sharper minima that generalize poorly (which I learned in 10-714). However, looking at the mean return, it seems like the network still benefits from a larger batch size. While the mean return increases with larger batch sizes, the high standard deviation suggests the results are not conclusive. It may come from poor generalization, but there is also the possibility that we are having bad luck when sampling i.i.d. views for training.

I also printed out and noticed that the maximum observation length of the Ant environment is 2000, which explains why the value stays unchanged after 2000. Furthermore, with the intention of fitting larger batches, I switched to other machines on Colab and discovered that the same seed does not guarantee the same behavior on different machines.

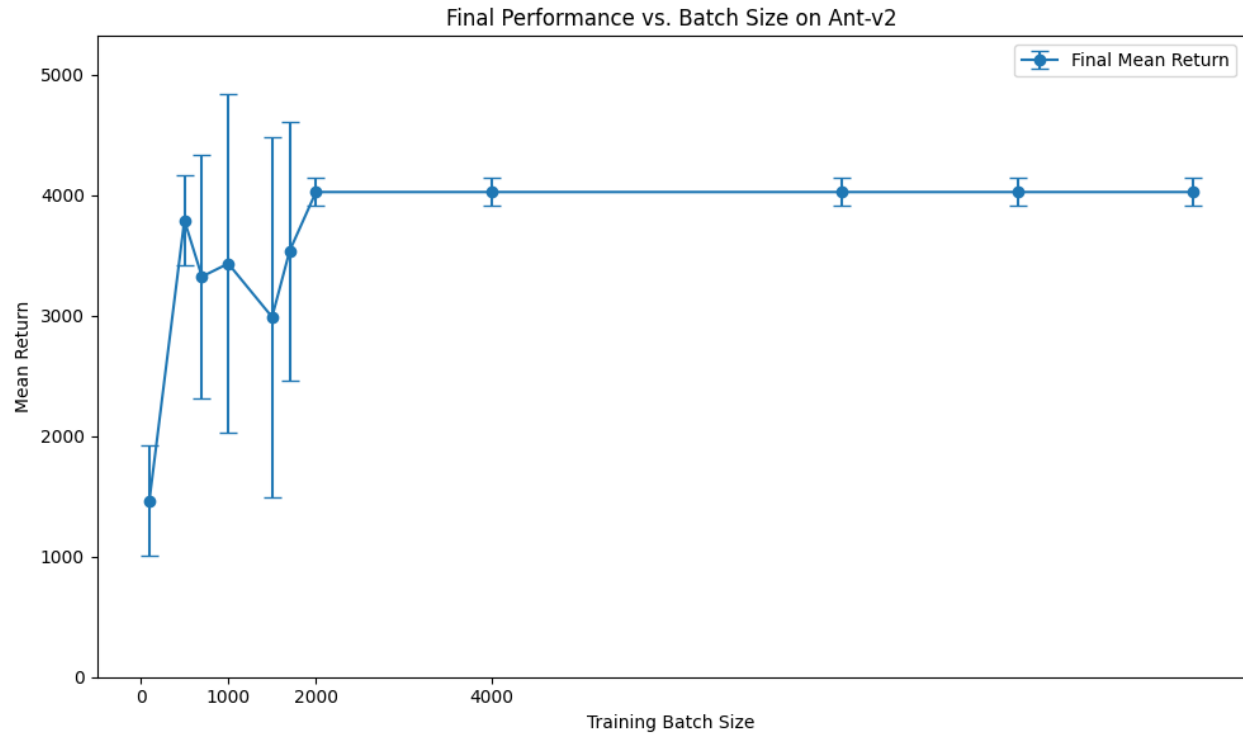


Figure 1: BC agent's performance varies with the training batch size parameter in the Ant environment. All others are kept the same as in Table 2.

2 DAgger (5.25 pt)

2.1 Part 2 (5.25 pt)

Fig. 2:

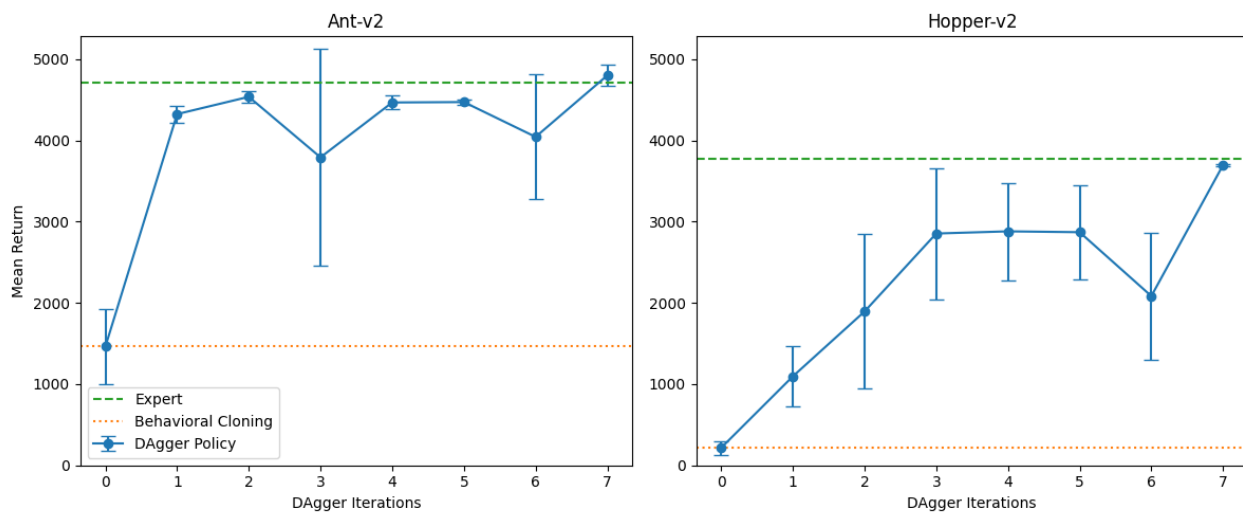
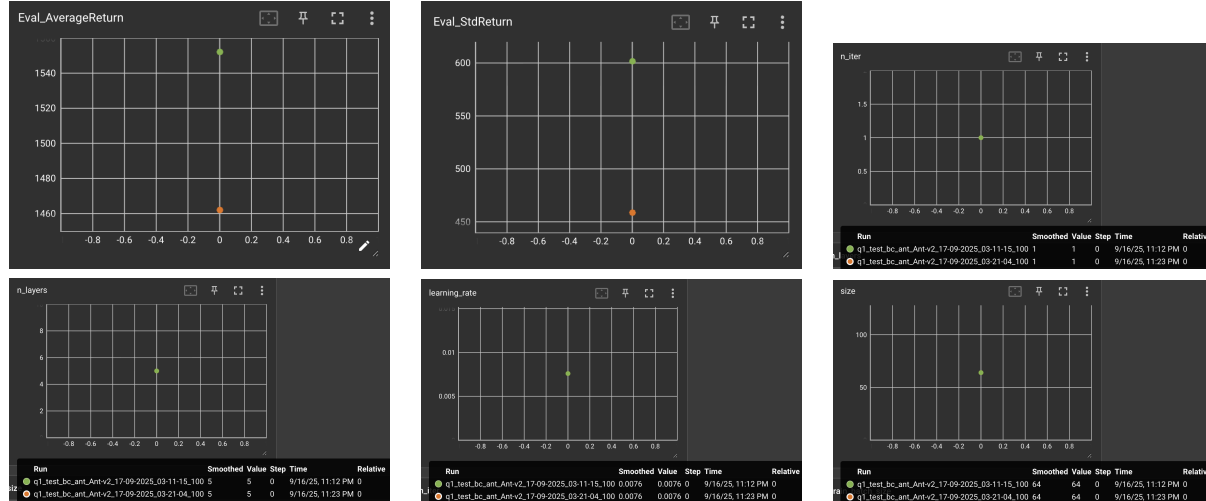


Figure 2: Learning curve, plotting the number of DAgger iterations vs. the policy's mean return, with error bars to show the standard deviation in Ant-v2 and Hopper-v2 environments. Parameters are the same as in Table 2.

Appendix A.

Same seed does not guarantee same behavior

I noticed that even with the same seed and all other parameters, the average and standard deviation differ from each other. The experiment was run on Colab CPU and Colab CPU (High-RAM).



Full parameters table

The following is the full parameters I used through out the assignment (except for Part 2).

Table 3: Parameters I used throughout the assignment, except for Part 2.

Parameter	BC	DAgger
Shared Parameters		
ep_len	1000	
save_params	False	
num_agent_train_steps_per_iter	1000	
batch_size	1000	
eval_batch_size	5000	
train_batch_size	100	
max_replay_buffer_size	1000000	
n_layers	5	
size	64	
learning_rate	7.6e-3	
video_log_freq	-1	
scalar_log_freq	1	
no_gpu	False	
which_gpu	0	
seed	1	
Variable Parameters		
do_dagger	False	True
n_iter	1	8